

DPCM Predictor Assignment

Part A — First-Order Prediction ($\hat{x}[n] = x[n-1]$)

Sequence: $x[n] = \{10, 12, 15, 17, 18, 20, 21, 23\}$

n	0	1	2	3	4	5	6	7
$x[n]$	10	12	15	17	18	20	21	23
$\hat{x}[n]$	-	10	12	15	17	18	20	21
$e[n]$	-	2	3	2	1	2	1	2

Range of $x[n]$: $23 - 10 = 13$. Range of $e[n]$: $3 - 1 = 2$. The prediction error occupies a much smaller dynamic range than the original signal, so it can be quantized with fewer bits for the same step size, or with a finer step size using the same number of bits.

Part B — Predictor Coefficient MSE Comparison Table

Sequence used: $x[n] = \{10, 12, 15, 17, 18, 20, 21, 23\}$. Predictor: $\hat{x}[n] = a \cdot x[n-1]$

Predictor coefficient, a	Prediction error $e[n]$	Prediction MSE
0	12, 15, 17, 18, 20, 21, 23	336.00
0.5	7, 9, 9.5, 9.5, 11, 11, 12.5	101.25
1	2, 3, 2, 1, 2, 1, 2	3.857
1.2	0, 0.6, -1, -2.4, -1.6, -3, -2.2	3.360

Among the tested fixed values, $a = 1.2$ gives the smallest MSE (3.360), slightly better than $a = 1$ (3.857). This shows that the true optimum lies near, but not exactly at, $a = 1$ for this particular sequence.

Part C — Optimum Predictor Coefficient

$$a_{\text{opt}} = (\sum x[n]x[n-1]) / (\sum x^2[n-1]) = 2124 / 1923 \approx 1.1045$$

Predictor	a	MSE
$a = 0$	0	336.00
$a = 0.5$	0.5	101.25
$a = 1$	1	3.857
$a = 1.2$	1.2	3.360
a_{opt} (Part C)	1.1045	0.856

The optimum coefficient $a_{\text{opt}} \approx 1.1045$ produces a dramatically lower MSE (0.856) than any of the fixed values tested in Part B, confirming that least-squares optimization of the predictor coefficient gives the best achievable first-order prediction for this sequence.

Part E — Predictor Coefficient MSE (MATLAB Signal)

Signal: $x[n] = 2\cos(0.05\pi n) + 0.5\cos(0.2\pi n)$, $n = 0, \dots, 199$. Predictor: $\hat{x}[n] = a \cdot x[n-1]$

a	0	0.5	0.8	1	1.2
MSE	2.1043	0.5740	0.1616	0.0974	0.2018

Among the tested values, $a = 1$ gives the minimum MSE (0.0974). Since this signal is a sum of slowly varying cosines, consecutive samples are strongly correlated and close to equal, so the simple predictor $\hat{x}[n] = x[n-1]$ ($a = 1$) already performs very well; $a = 1.2$ overshoots and increases the error again.

Part F — Optimum Predictor Coefficient (MATLAB Signal)

$a_{\text{opt}} = \Sigma(x[n]x[n-1]) / \Sigma(x^2[n-1]) \approx 0.9762$ (computed over $n = 1, \dots, 199$)

Predictor	a	MSE
a = 1 (simple predictor)	1	0.09741
a _{opt} (Part F)	0.9762	0.09621

The difference between MSE at $a = 1$ (0.09741) and MSE at a_{opt} (0.09621) is very small (about 1.2%). This is because the signal is smooth and highly correlated (dominated by a slowly varying cosine component), so the simple predictor $a = 1$ is already very close to optimal; the least-squares correction offers only a marginal improvement for this type of signal.

Bonus — Second-Order Predictor Comparison

Predictor: $\hat{x}[n] = a_1x[n-1] + a_2x[n-2]$, with $a_1 \approx 1.4589$, $a_2 \approx -0.3995$ (verified in MATLAB).

Predictor	Coefficients	MSE
1st-order optimum ($n = 3:N$)	$a_{\text{opt}} = 1.1045$	0.8465
2nd-order optimum	$a_1 = 1.4589$, $a_2 = -0.3995$	0.7097

The second-order predictor reduces the MSE from 0.8465 (first-order optimum) to 0.7097, an improvement of about 16%, confirming that adding one more past sample can further exploit the correlation present in this sequence, though with added computational cost.

Final Conclusion

In a DPCM system, the predictor uses previously reconstructed samples to estimate the value of the current sample before it is transmitted, so that only the difference between the actual and predicted value — the prediction error $e[n]$ — needs to be quantized and sent. Because real-world signals such as speech are highly correlated from sample to sample, a well-designed predictor can estimate the next sample quite accurately, which makes $e[n]$ much smaller in amplitude and dynamic range than the original signal $x[n]$. This smaller dynamic range means fewer quantization levels, and therefore fewer bits, are needed to represent $e[n]$ at the same quality as directly quantizing $x[n]$. A good predictor, whether chosen empirically or derived through least-squares optimization as in Part C, minimizes the prediction error variance and therefore maximizes the prediction gain, directly improving the coding efficiency of the DPCM system without sacrificing reconstruction quality.